

Applied Class 1 – Solutions

Part A

This is a short activity to check you know the important ideas about logarithms. Without using a calculator (answers should have no decimal places), simplify the following as much as possible:

1. $\log_{10} 10 = 1$

2. $\log_{10} 1000 = 3$

3. $\log_{10} 1 = 0$

4. $\log_{10} 0.01 = \log_{10} 10^{-2} = -2$

5. $\log_{10}(10^x) = x$

6. $\log_{10} 200 = \log_{10}(100 \times 2) = \log_{10} 100 + \log_{10} 2 = 2 + \log_{10} 2$

7. $\log_{10} 0$ is undefined

8. $\log_2 16 = 4$

9. $\log_2 \frac{1}{4} = -2$

10. $\log_e 2.718282 \approx 1$

Part B

See “Randomisation Test Activity”.

Part C

A high-fat diet (HFD) is known to have a negative effect on bone quality in young and old people. A study was undertaken to evaluate the effects of an HFD on bone healing in growing female rats. Twenty-six postweaning female rats were randomly assigned to two groups (13 per group): a standard diet (SD) group and an HFD (with 60% of energy from fat) group. The rats were maintained under standard laboratory conditions at a temperature of 22 ± 2 °C and humidity of 55 ± 5 % with a 12-hour light and dark cycle and with free access to water and food. The rats received the assigned diets for 5 weeks, and in the third week they were submitted to an osteotomy procedure (i.e. bone cut) of the left tibia. At the end of the 5 weeks, the rats were euthanised and the tibiae were removed for mechanical testing. The mechanical testing involved a force being applied to the tibia at a speed of 1.0 mm/min to generate a tension load on the osteotomy site. The maximal load (N) and stiffness (N/mm) were measured. Lower maximal load ($p = 0.040$) and bending stiffness ($p = 0.020$) were observed in tibiae of HFD rats compared to SD rats.

1. What type of variable is *maximal load*? **Continuous**
2. What type of variable is *diet*? **Ordinal (diet ordered by fat content). Nominal would be an acceptable answer.**
3. Which variable is the response variable? **Maximal load**
4. What type of study is this? **Randomised comparative experiment**
5. What would be a suitable null hypothesis associated with the p -value for maximal load?
 H_0 : The level of dietary fat has no effect on the average bone maximal load following osteotomy in rats
6. What are we able to conclude about the effect of diet on bone maximal load following osteotomy? **There is moderate evidence against the null hypothesis ($p = 0.04$), suggesting that high fat diets in rats reduces the average bone maximal load**

Reference

J.S Yamanaka, G.R. Yanagihara, B.L. Carlos, J. Ramos, B.B. Brancalion, A.P. Macedo, J.P.M Issa & A.C. Shimano (2018) A high-fat diet can affect bone healing in growing rats, *Journal of Bone and Mineral Metabolism*, 36, 255-263. doi: 10.1007/s00774-017-0837-4

Part D

How long is a piece of string?